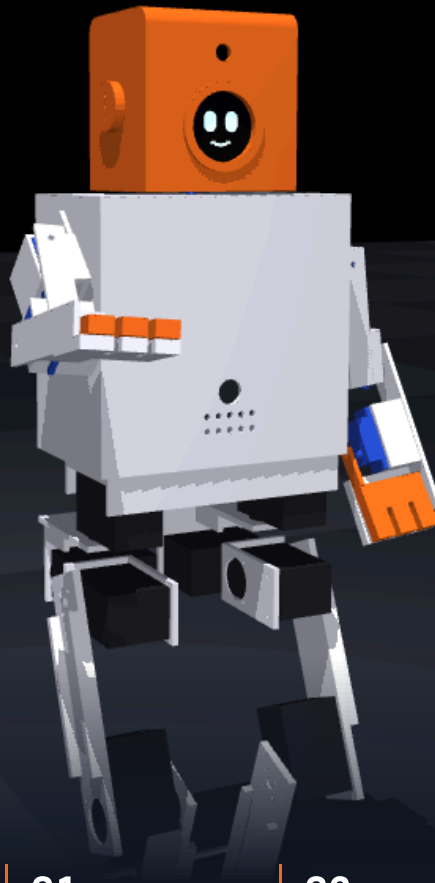


JX0

walks · talks · shows a face · uses its hands



51.7 cm

HEIGHT

2.07 kg

MASS

21

SERVOS

23

PRINTED PARTS

27

BOM LINES

PARTS FOR ONE ROBOT

₹47,106

UNDER THE ₹50,000 BUDGET BY

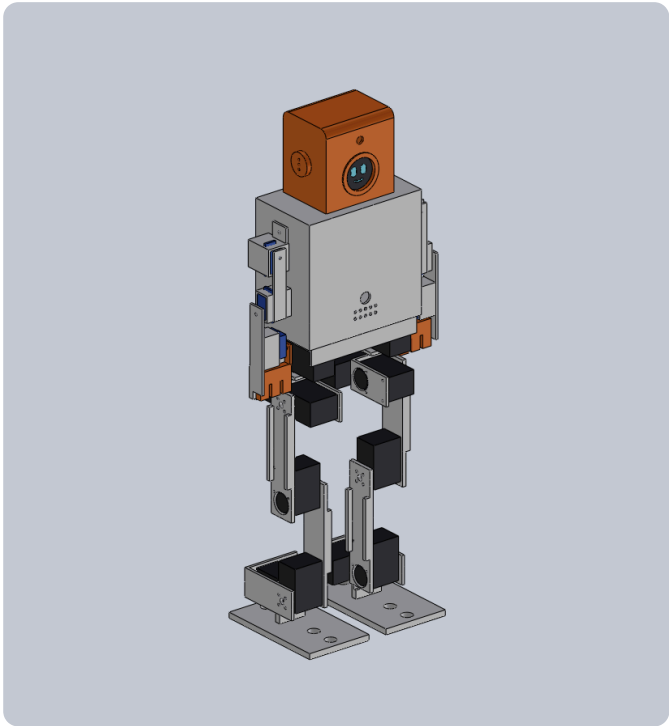
₹2,894

1. What JX0 is

JX0 is a small humanoid robot designed to be built by one student at home, from a 3D printer and parts that can be ordered online in India. It is the working prototype of the JX1 project: it uses the same design pipeline (gait planner, servo sizing, physics simulation) at a price a student can afford.

Specification

Height / mass	51.7 cm / 2.07 kg (CAD, all parts)
Legs	6 joints each: Feetech ST3215-C018 12 V serial bus servos (30 kg-cm stall, position feedback)
Arms and hands	shoulder pitch and roll, elbow and a three-finger gripper per arm: MG90S micro servos
Head	neck yaw, a 1.28-inch round screen as the face (eyes and a mouth that moves with speech), camera
Brain	Raspberry Pi 4: 50 Hz gait playback, IMU balance, offline speech recognition and voice
Talking	Claude over Wi-Fi writes the replies and calls the robot's actions (wave, walk, take, give)
Power	3S 2200 mAh LiPo, about 20-30 minutes of walking (ESTIMATED)
Structure	PETG on a home 3D printer: 23 parts, 379 g



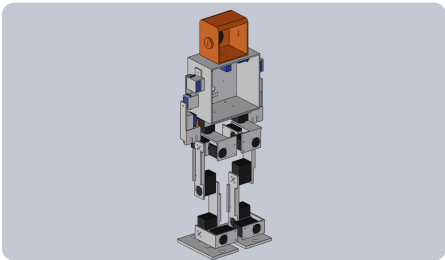
Done (on the computer)

- ✓ SolidWorks CAD: 23 printable parts, 46-component assembly, built by script
- ✓ Servo sizing for every leg joint: all pass (tightest margin 1.89×)
- ✓ Walking in a physics simulation of the CAD robot: 14 of 14 gaits pass
- ✓ Robot program (voice, Claude, face, walking, hands) runs on the simulated robot
- Next: buy the parts → print → assemble → first steps

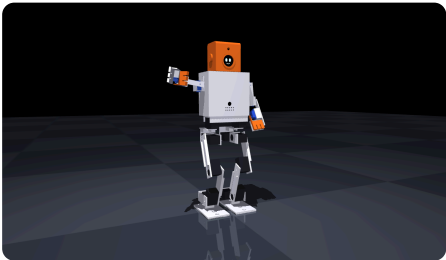
How this estimate was made

- Every line is a real product page at an Indian store (Robu, Evelta, Robocraze, ThinkRobotics, Quartz Components and others), price checked September 2026.
- GST is included where the store states it; shipping is not included.
- Two small lines (button, jumper wires) are ESTIMATED.

From CAD to a working robot



SolidWorks assembly, back view



waving (simulation, the robot's own program)



walking at 0.067 m/s (simulation)

2. Cost summary

PARTS TOTAL

₹47,106

27 line items, one robot

LEG SERVOS

62 %

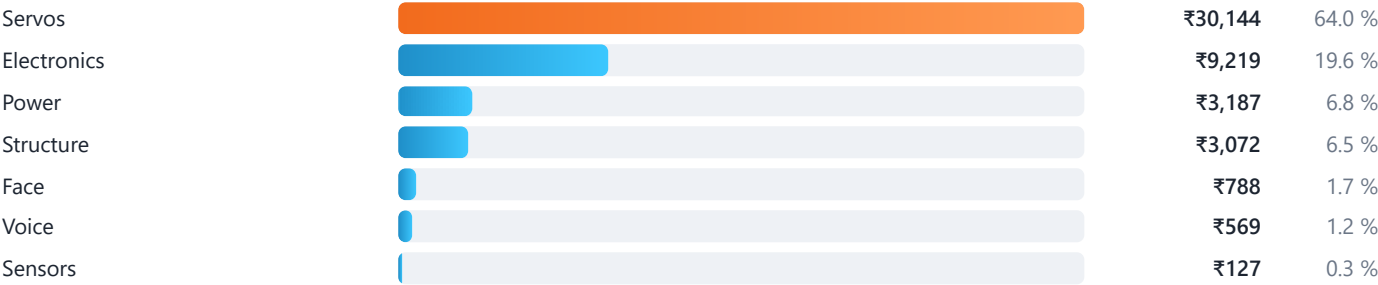
of the cost: 12 × ST3215 at ₹2,419

LEFT IN A ₹50K BUDGET

₹2,894

for shipping and spares

By group



Groups

GROUP	WHAT	₹	SHARE
Servos	12 × ST3215 leg servos, 9 × MG90S	₹30,144	64.0 %
Electronics	Raspberry Pi 4, microSD, servo driver, wiring	₹9,219	19.6 %
Power	LiPo, charger, converters, switch, alarm	₹3,187	6.8 %
Structure	PETG filament, screws, inserts, foot rubber	₹3,072	6.5 %
Face	round display, camera	₹788	1.7 %
Voice	microphone, amplifier, speaker	₹569	1.2 %
Sensors	IMU	₹127	0.3 %
Total		₹47,106	100 %

Five biggest lines

ITEM	MODEL	QTY	₹
Leg servo (12 per robot)	Feetech ST3215-C018 12 V 30 kg-cm serial bus servo	12	₹29,028
Brain	Raspberry Pi 4 Model B 2 GB (with case)	1	₹7,000
Battery	Bonka 3S 11.1 V 2200 mAh 35C LiPo	1	₹2,039
microSD card	SanDisk Ultra 32 GB A1	1	₹1,499
PETG filament	Sunlu PETG 1.75 mm 1 kg	2	₹1,478

Why these leg servos: the walking needs up to 1.56 N·m at the hip (with a 1.5× safety margin), which is more than MG996R-class hobby servos give. The serial bus servos also report their position, which the calibration and the balance loop use.

3. Complete list of items

#	ITEM	STORE	QTY	UNIT	TOTAL
01	Leg servo (12 per robot) Feetech ST3215-C018 12 V 30 kg-cm serial bus servo	Evelta	12	₹2,419	₹29,028
02	Arm, gripper and neck servo TowerPro MG90S metal-gear micro servo	Quartz Components	9	₹124	₹1,116
03	Bus servo driver Waveshare Serial Bus Servo Driver board	ThinkRobotics	1	₹500	₹500
04	Brain Raspberry Pi 4 Model B 2 GB (with case)	ThinkRobotics	1	₹7,000	₹7,000
05	microSD card SanDisk Ultra 32 GB A1	Quartz Components	1	₹1,499	₹1,499
06	IMU MPU6050 gyro + accelerometer	Quartz Components	1	₹127	₹127
07	Microphone INMP441 I2S MEMS microphone	Quartz Components	1	₹145	₹145
08	Amplifier MAX98357A I2S 3 W class-D amplifier	Quartz Components	1	₹124	₹124
09	Speaker Waveshare 2030 cavity speaker 8 Ω 2 W	ThinkRobotics	1	₹300	₹300
10	Round display (face) 1.28 inch round TFT 240x240 GC9A01 SPI	Robu	1	₹469	₹469
11	Camera OV5647 5 MP Raspberry Pi camera Rev 1.3	Robu	1	₹319	₹319
12	Battery Bonka 3S 11.1 V 2200 mAh 35C LiPo	Robocraze	1	₹2,039	₹2,039
13	Charger B3 balance charger for 2S/3S LiPo	Quartz Components	1	₹294	₹294
14	5 V supply for the Pi UBEC 5 V 5 A (2-8S input)	Robu	1	₹409	₹409
15	5 V supply for the micro servos MINI560 5 V 5 A buck (6-24 V in)	Quartz Components	1	₹65	₹65
16	Battery connector XT60 male/female pair	Robocraze	2	₹29	₹58
17	Power switch 10 A DC-rated toggle switch (CALONIX)	Quartz Components	1	₹63	₹63
18	Power wire 16 AWG silicone wire (per metre)	Robu	2	₹49	₹98
19	Signal wire 20 AWG silicone wire (per metre)	Robu	3	₹21	₹63
20	Battery alarm 1-8S LiPo voltage checker / low-voltage buzzer	Quartz Components	1	₹98	₹98
21	PETG filament Sunlu PETG 1.75 mm 1 kg	Robu	2	₹739	₹1,478
22	M2 x 6 screws (horns and servo cases) M2 x 6 12.9 socket head (pack of 10)	Indian Hobby Center	10	₹49	₹490
23	M3 x 8 screws (torso to pelvis) M3 x 8 socket head SS202 (pack of 10)	Quartz Components	1	₹35	₹35
24	M3 heat-set inserts M3 x 4 brass heat-set inserts (25 pcs)	Robu	1	₹91	₹91
25	Foot grip Natural rubber sheet 1 mm x 100 mm x 1 m	IndustryBuying	1	₹978	₹978
26	Push-to-talk button 12 mm momentary push button	any electronics shop	1	₹20	₹20
27	Jumper wires and servo extensions Dupont jumpers + 3-pin extensions	any electronics shop	1	₹200	₹200

Total for one JX0

₹47,106

Rupees Forty-Seven Thousand One Hundred Six only